Optimized Placement of Flood Sensors to Monitor Road Waterlogging in Urban Areas: Gurugram

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1. Introduction

This project focuses on the spatial optimization of sensor placement to improve the monitoring of road waterlogging during rainfall events.

In many Indian cities, roads get flooded even during moderate rainfall. Water accumulates on roads because the drainage system is unable to carry the runoff efficiently, or because low-lying areas collect water faster than it can drain. This causes traffic congestion, delays for emergency vehicles, and inconvenience to daily commuters.

Although flood sensors can help detect such situations in real time, most cities have very few sensors. Even when sensors exist, they are often placed without a clear planning logic. As a result, some important flood-prone roads are not monitored, while other locations collect repeated or less useful information. This project addresses this gap by treating sensor placement as a planning and optimization problem.

2. Problem Statement

Road waterlogging is a spatially uneven problem. Some locations flood repeatedly, while others remain unaffected. However, there is no systematic method to decide where flood sensors should be placed to capture these patterns.

Since sensors cannot be installed everywhere, poor placement can lead to:

- Flood-prone roads remaining unmonitored
- Inefficient use of available sensors
- Delayed response during heavy rainfall

There is a need for a structured, network-based approach to decide sensor locations so that flooding can be observed effectively with limited resources.

3. Motivation

Urban flooding is not random. It follows the structure of the city itself. Water flows along roads, accumulates in low-lying areas, and finally moves through the drainage network towards rivers. This means that flooding is strongly controlled by spatial layout, connectivity, and elevation.

However, sensor placement decisions are often made without considering these factors. Sensors are placed where it is convenient, rather than where they are most useful. This motivated the idea that flood monitoring should be planned in the same way as other infrastructure systems.

The key idea of this project is simple: if the drainage network is treated as a system, then sensors should be placed by understanding how information flows through that system. By using concepts from infrastructure systems planning, networks, and optimization, sensor placement can be improved without increasing the number of sensors.

4. Objectives

The objectives of this project are:

- To model the urban drainage system as a connected network
- To identify important locations in the network that influence flood observability
- To formulate sensor placement as a spatial optimization problem
- To determine the minimum number of sensors required for effective monitoring
- To develop a planning framework that can be applied to other cities

5. Methodology

The methodology treats flood sensor placement as a network-based planning and optimization problem. Instead of focusing on detailed hydraulic simulations, the approach concentrates on how information about flooding propagates through the drainage network and how sensors can be placed to observe this information efficiently.

The methodology integrates drainage network modeling, data-driven importance weighting, efficiency screening, and optimization to determine effective sensor locations under practical constraints.

5.1 Methodology Map

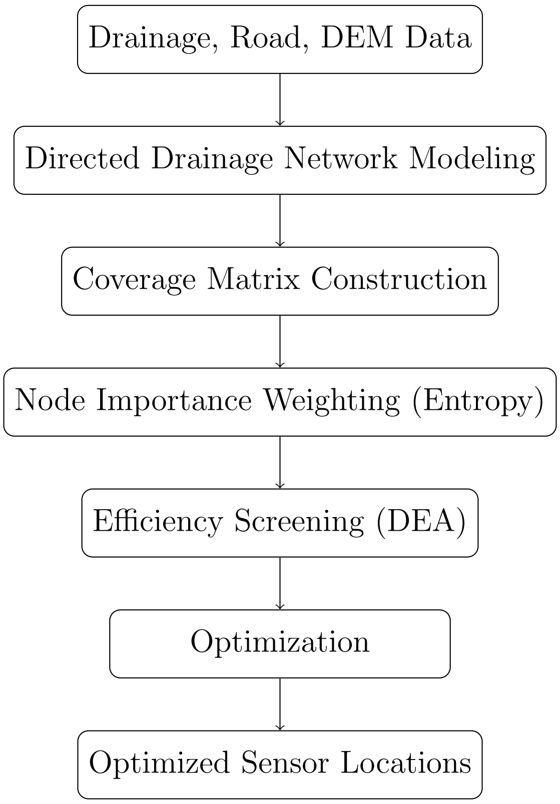

Figure 1: Overall methodology for flood sensor placement

5.2 Study Area and Data

The study area is Gurugram. The following datasets are used:

- Urban drainage or sewer network data (pipes and manholes). If unavailable, sewer lines are assumed to follow road alignments.
- Drainage outfall locations connected to rivers.
- Digital Elevation Model (DEM) to identify low-lying areas and infer flow direction.
- Road network data to represent traffic importance.

5.3 Drainage Network Modeling

The drainage system is represented as a directed network:

- Nodes represent manholes, junctions, and river outfalls.
- Edges represent drainage pipes.
- Flow direction is assigned using elevation and network connectivity.

River outfalls are treated as terminal nodes, since flooding information from upstream locations ultimately propagates towards these points.

5.4 Candidate Sensor Locations

Sensors are allowed only at physically feasible locations such as:

- Manholes,
- Major drainage junctions,
- Locations near river outfalls.

These feasible locations form the candidate sensor set V_s.

5.5 Coverage and Observability

When flooding occurs upstream, its effects propagate through the drainage network. As a result, a sensor placed at a downstream location can provide information about flooding conditions upstream.

To mathematically represent this idea, a coverage matrix c_{ij} is defined:

$$c_{ij} = \begin{cases} 1, & \text{if a sensor placed at node } i \text{ can observe node } j, \\ 0, & \text{otherwise.} \end{cases}$$

The value of c_{ij} is determined based on:

- Directed upstream–downstream connectivity in the drainage network,
- A maximum allowable network distance $D_{\max}$, and
- Hydraulic direction, allowing only upstream observability.

This matrix converts physical drainage behavior into a mathematical structure that links sensor placement decisions to network observability.

5.6 Node Importance Weighting (Entropy-Based)

Flooding at different locations does not have the same consequence. Therefore, each node $j \in V$ is assigned an importance weight $w_j \in [0, 1]$, representing the relative impact if flooding occurs at that node.

Step 1: Decision Matrix Construction

Let there be n nodes and m criteria. A decision matrix

$$\mathbf{Z} = [z_{jk}]$$

is constructed, where

$$z_{jk} = \text{value of criterion } k \text{ at node } j.$$

The criteria considered include:

- Low-lying index derived from DEM,
- Traffic importance,
- Proximity to critical infrastructure,
- Proximity to river outfalls (computed using network distance).

Step 2: Normalization

Since the criteria are measured in different units, each column of the decision matrix is normalized as:

$$p_{jk} = \frac{z_{jk}}{\sum_{j=1}^{n} z_{jk}},$$

which ensures:

$$\sum_{j=1}^{n} p_{jk} = 1 \quad \forall k.$$

Step 3: Entropy Calculation

The entropy value of criterion k is computed as:

$$E_k = -\frac{1}{\ln(n)} \sum_{j=1}^{n} p_{jk} \ln(p_{jk}).$$

Entropy measures the degree of uniformity of a criterion across the network. If a criterion has similar values at most nodes, it exhibits high entropy and provides limited information. In contrast, criteria with large spatial variation have lower entropy and carry more useful information for distinguishing flood-prone locations.

Step 4: Degree of Diversification

The information contribution of each criterion is given by:

$$d_k = 1 - E_k.$$

A higher value of d_k indicates that criterion k provides greater discriminatory power.

Step 5: Criterion Weights

Entropy-based criterion weights are computed as:

$$w_k = \frac{d_k}{\sum_{k=1}^{m} d_k},$$

which satisfies:

$$\sum_{k=1}^{m} w_k = 1.$$

Step 6: Node Importance Score

Finally, the importance of each node is computed as a weighted sum:

$$w_j = \sum_{k=1}^{m} w_k \, z_{jk}.$$

Nodes that score high across multiple important criteria receive higher importance values.

5.7 Efficiency Screening Using Data Envelopment Analysis (DEA)

Data Envelopment Analysis (DEA) is used to evaluate the relative efficiency of candidate sensor locations before final optimization.

Decision-Making Units

Each candidate sensor location $i \in V_s$ is treated as a decision-making unit (DMU).

Inputs and Outputs

For each DMU i, the following inputs and outputs are defined:

Inputs (to be minimized):

- x_{i1}: redundancy with nearby sensors,
- x_{i2}: overlap in coverage.

Outputs (to be maximized):

- y_{i1}: number of nodes covered,
- y_{i2}: total importance covered.

DEA Efficiency Model

The efficiency score of sensor location i is computed as:

$$\max \quad \frac{\sum_r u_r y_{ir}}{\sum_q v_q x_{iq}}$$

subject to:

$$\frac{\sum_r u_r y_{jr}}{\sum_q v_q x_{jq}} \leq 1 \quad \forall j,$$
$$u_r \geq 0, \quad v_q \geq 0,$$

where u_r and v_q are weights assigned to outputs and inputs, respectively. These weights are chosen to maximize the efficiency of DMU i.

Interpretation

An efficiency score equal to 1 indicates an efficient sensor location, while a score less than 1 indicates inefficiency. DEA identifies sensor locations that provide high coverage with minimal redundancy. Inefficient locations can be filtered or ranked lower before optimization.

5.8 Optimization Formulation (Multi-Objective Perspective)

The final sensor placement problem is formulated as a binary integer optimization problem.

Decision Variables

$$x_i = \begin{cases} 1, & \text{if a sensor is placed at node } i, \\ 0, & \text{otherwise,} \end{cases} \qquad \forall i \in V_s$$

Additionally, define:

$$y_j = \begin{cases} 1, & \text{if node } j \text{ is covered,} \\ 0, & \text{otherwise,} \end{cases} \qquad \forall j \in V$$

Objective Function

The objective is to maximize the total importance of covered nodes:

$$\max \sum_{j \in V} w_j \, y_j.$$

Constraints

(a) Coverage Constraint

$$\sum_{i \in V_s} c_{ij} x_i \geq y_j \quad \forall j \in V$$

(b) Directional Constraint

$$c_{ij} = 0 \quad \text{if node } j \text{ is downstream of node } i$$

(c) Maximum Influence Distance Constraint

$$c_{ij} = 0 \quad \text{if } d_{ij} > D_{\max}$$

(d) River Outfall Constraint

$$\sum_{i \in V_{\text{outfall}}} x_i \geq 1$$

(e) Sensor Availability Constraint

$$\sum_{i \in V_s} x_i \leq K$$

(f) Binary Constraints

$$x_i \in \{0, 1\}, \quad y_j \in \{0, 1\}$$

Solver and Implementation

The resulting formulation is a binary integer programming problem and can be solved using standard mixed-integer linear programming solvers such as CPLEX which is used in MILP.

6. Expected Outcomes

The expected outcomes of this project are:

- An optimized map of flood sensor locations
- Improved monitoring of flood-prone road segments
- Reduction in redundant or unnecessary sensors
- A planning-oriented framework for urban flood monitoring